

Lab 1

SELECT rental_id, customer_id, return_date FROM rental WHERE rental_id = 2;

The screenshot shows the PostgreSQL IDE interface. On the left, the 'Object Explorer' pane displays the database structure for 'PostgreSQL 16', including 'dvdrental' and its tables. The main query editor contains the following SQL statement:

```
1 SELECT rental_id, customer_id, return_date FROM rental WHERE rental_id = 2;
```

The 'Data Output' pane at the bottom shows the results of the query:

rentalId [PK] integer	customer_id smallint	return_date timestamp without time zone
1	2	2005-05-28 19:40:33

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.105'.

BEGIN;

UPDATE rental SET return_date = NOW() WHERE rental_id = 2;

INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');

COMMIT;

The screenshot shows the PostgreSQL IDE interface with a multi-line SQL query in the editor:

```
1 SELECT rental_id, customer_id, return_date FROM rental WHERE rental_id = 2;
2 BEGIN;
3
4 UPDATE rental SET return_date = NOW() WHERE rental_id = 2;
5
6 INSERT INTO payment (customer_id, staff_id, rental_id, amount, payment_date)
7 VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
8
9 COMMIT;
```

The 'Messages' pane at the bottom displays an error message:

```
ERROR: invalid input syntax for type timestamp: "WRONG_DATE_FORMAT"
LINE 7: VALUES (459, 1, 2, 99.99, 'WRONG_DATE_FORMAT');
                                     ^
SQL state: 22007
Character: 248
```

The status bar at the bottom indicates 'Total rows: 1' and 'Query complete 00:00:00.112'.

Object Explorer | Dashboard | dvdrental/postgres@PostgreSQL 16

Servers

- PostgreSQL 16
 - Databases (2)
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 - public
 - Tables (15)
 - actor
 - address
 - category
 - city
 - country
 - customer
 - columns
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 - indexes

Query: `SELECT rental_id, customer_id, return_date FROM rental WHERE rental_id = 2; ROLLBACK; SELECT return_date FROM rental WHERE rental_id = 2;`

Data Output: Showing rows: 1 to 1 of 1

	return_date timestamp without time zone
1	2005-05-28 19:40:33

Successfully run. Total query runtime: 85 msec. 1 rows affected.

Total rows: 1 Query complete 00:00:00.085

Lab prep

Object Explorer

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Query

```

1 BEGIN;
2 UPDATE film SET rental_rate = 9.99
3 WHERE film_id IN (SELECT film_id FROM film_category WHERE category_id = 1);
4
5 COMMIT;

```

Data Output

WARNING: there is already a transaction in progress
COMMIT

Query returned successfully in 49 msec.

Total rows: Query complete 00:00:00.049

LF Ln 5, Col 8

Query

```

1 SELECT title, rental_rate FROM film
2 WHERE film_id IN (SELECT film_id FROM film_category WHERE category_id = 1);

```

Data Output

Showing rows: 1 to 64 Page No: 1 of 1

	title character varying (255)	rental_rate numeric (4,2)
1	Casualties Encino	9.99
2	Uprising Uptown	9.99
3	Amadeus Holy	9.99
4	American Circus	9.99
5	Antitrust Tomatoes	9.99
6	Ark Ridgemont	9.99
7	Barefoot Manchurian	9.99
8	Berets Agent	9.99
9	Bride Intrigue	9.99
10	Bull Shawshank	9.99
11	Caddyshack Jedi	9.99
12	Campus Remember	9.99

Total rows: 64 Query complete 00:00:00.219

LF Ln 2, Col 76

Lab: Consistency

BEGIN;

UPDATE film SET replacement_cost = -5.00

WHERE title = 'Academy Dinosaur';

COMMIT;

Object Explorer: PostgreSQL 16, dvrental/postgres@PostgreSQL 16

Query:

```

1 BEGIN;
2 UPDATE film SET replacement_cost = -5.00
3 WHERE title = 'Academy Dinosaur';
4 COMMIT;

```

Data Output: Messages

ERROR: new row for relation "film" violates check constraint "min_replacement_cost"
Failing row contains (1, Academy Dinosaur, A Epic Drama of a Feminist And a Mad Scientist who must Battle a..., 2006, 1, 6, 0.99, 86, -5.00, PG, 2026-09-27 18:29:47.002958, {"Deleted Scenes","Behind the Scenes"}, 'academ':1 'battl':15 'canadian':20 'dinosaur':2 'drama':5 'epi...).

SQL state: 23514
Detail: Failing row contains (1, Academy Dinosaur, A Epic Drama of a Feminist And a Mad Scientist who must Battle a..., 2006, 1, 6, 0.99, 86, -5.00, PG, 2026-09-27 18:29:47.002958, {"Deleted Scenes","Behind the Scenes"}, 'academ':1 'battl':15 'canadian':20 'dinosaur':2 'drama':5 'epi...).

Total rows: 3 Query complete 00:00:00.060

Object Explorer: PostgreSQL 16, dvrental/postgres@PostgreSQL 16

Query:

```

1 SELECT film_id, title, replacement_cost FROM film WHERE title = 'Academy Dinosaur';

```

Data Output: Messages

film_id [PK] integer	title character varying (255)	replacement_cost numeric (5,2)
1	Academy Dinosaur	20.99

Showing rows: 1 to 1 of 1 Page No: 1

Total rows: 1 Query complete 00:00:00.061

Exercise 1: Không. Vì bước INSERT payment bị lỗi (sai định dạng ngày), toàn bộ transaction bị đánh dấu "failed", nên khi COMMIT chạy tới thì không có gì được lưu kể cả UPDATE ở bước 1 đã "chạy được" trước đó.

Exercise 2

```
SELECT column_name, data_type, is_nullable  
FROM information_schema.columns  
WHERE table_name = 'staff';
```

The screenshot shows a PostgreSQL client interface with a query editor and a results pane. The query editor contains the following SQL:

```
1 SELECT column_name, data_type, is_nullable  
2 FROM information_schema.columns  
3 WHERE table_name = 'staff';
```

The results pane displays the following table:

	column_name name	data_type character varying	is_nullable character varying (3)
1	staff_id	integer	NO
2	first_name	character varying	NO
3	last_name	character varying	NO
4	address_id	smallint	NO
5	email	character varying	YES
6	store_id	smallint	NO
7	active	boolean	NO
8	username	character varying	NO
9	password	character varying	YES
10	last_update	timestamp without time zo...	NO
11	picture	bytea	YES

At the bottom of the interface, a status bar indicates: "Total rows: 11 Query complete 00:00:00.073". A green notification box at the bottom right says: "✓ Successfully run. Total query runtime: 73 msec. 11 rows affected. X".

```
BEGIN;
```

```
INSERT INTO staff (first_name, last_name, address_id, email, store_id, username,  
active)  
VALUES ('Test', 'Staff', 1, 'test@edu.vn', 1, 'teststaff', true);
```

```
SAVEPOINT staff_created;
```

```
-- Cố tình gán store_id không tồn tại (giả sử chỉ có store_id 1 và 2)
```

```
UPDATE staff SET store_id = 999 WHERE username = 'teststaff';
```

Object Explorer

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Dashboard X dvdrantal/postgres@PostgreSQL 16+ X

dvdrantal/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```
1 SELECT column_name, data_type, is_nullable
2 FROM information_schema.columns
3 WHERE table_name = 'staff';
4
5 BEGIN;
6
7 INSERT INTO staff (first_name, last_name, address_id, email, store_id, username, active)
8 VALUES ('Test', 'Staff', 1, 'test@edu.vn', 1, 'teststaff', true);
9
10 SAVEPOINT staff_created;
11
12 -- Có tình gán store_id không tồn tại (giả sử chỉ có store_id 1 và 2)
13 UPDATE staff SET store_id = 999 WHERE username = 'teststaff';
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 60 msec.

Total rows: Query complete 00:00:00.060 LF Ln 13, Col 62

ROLLBACK TO staff_created;
UPDATE staff SET store_id = 1 WHERE username = 'teststaff';
COMMIT;

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Dashboard X dvdrantal/postgres@PostgreSQL 16+ X

dvdrantal/postgres@PostgreSQL 16

Query Query History Scratch Pad X

```
1 SELECT column_name, data_type, is_nullable
2 FROM information_schema.columns
3 WHERE table_name = 'staff';
4
5 BEGIN;
6
7 INSERT INTO staff (first_name, last_name, address_id, email, store_id, username, active)
8 VALUES ('Test', 'Staff', 1, 'test@edu.vn', 1, 'teststaff', true);
9
10 SAVEPOINT staff_created;
11
12 -- Có tình gán store_id không tồn tại (giả sử chỉ có store_id 1 và 2)
13 UPDATE staff SET store_id = 999 WHERE username = 'teststaff';
```

Data Output Messages Notifications

UPDATE 1

Query returned successfully in 60 msec.

Total rows: Query complete 00:00:00.060 LF Ln 13, Col 62

SELECT staff_id, first_name, last_name, store_id FROM staff WHERE username = 'teststaff';

The screenshot shows a PostgreSQL IDE interface. On the left is the 'Object Explorer' pane showing a tree view of the database structure, including 'PostgreSQL 16', 'Databases (2)', 'dvdrental', and 'public' schema. The main area is divided into a 'Query' editor and a 'Data Output' pane. The 'Query' editor contains the following SQL script:

```
BEGIN;  
INSERT INTO staff (first_name, last_name, address_id, email, store_id, username, active)  
VALUES ('Test', 'Staff', 1, 'test@edu.vn', 1, 'teststaff', true);  
SAVEPOINT staff_created;  
  
-- Có tính gán store_id không tồn tại (giả sử chỉ có store_id 1 và 2)  
UPDATE staff SET store_id = 999 WHERE username = 'teststaff';  
ROLLBACK TO staff_created;  
  
UPDATE staff SET store_id = 1 WHERE username = 'teststaff';  
COMMIT;  
  
SELECT staff_id, first_name, last_name, store_id FROM staff WHERE username = 'teststaff';
```

The 'Data Output' pane shows the results of the final SELECT query. It displays a table with 4 columns: staff_id, first_name, last_name, and store_id. The results are as follows:

staff_id	first_name	last_name	store_id
1	3	Test	Staff
2	4	Test	Staff
3	5	Test	Staff

At the bottom of the IDE, a status bar indicates 'Total rows: 3' and 'Query complete 00:00:00.061'.